# Annie Prasanna Manoharan

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## Education

**University of Arkansas** Aug 2024 – May 2026
*Master of Science in Computer Engineering* *Fayetteville, AR*
**Coursework:** Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

**Sri Ramakrishna Engineering College** Aug 2019 – May 2023
*Bachelor of Engineering in Electronics and Communication* *India*

## Technical Skills

**Data Engineering:** Flink SQL, Apache Kafka, Apache Flink, Apache Spark, Apache Airflow, ETL/ELT, ClickHouse
**Cloud and Tools:** Data pipeline design, Apache Iceberg, Schema evolution, Distributed data systems, Amazon S3, Stream processing
**Languages:** SQL, Python
**Machine Learning and AI:**

## Professional Experience

**Data Engineering Intern** Jul 2024 – Present
*InsOps Inc.* *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

**Teaching Assistant** Sep 2024 – May 2026
*University of Arkansas, Dept. of Computer Science and Engineering* *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

**Research Assistant** Sep 2024 – Jan 2026
*AICV Lab, University of Arkansas* *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

**Project Engineer** Jun 2023 – Jun 2024
*Soliton Technologies* *India*

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements

## Projects

**Kafka-to-Lakehouse Ingestion Connector** *Apache Kafka, Python, Amazon S3, Apache Iceberg*

- Built a configurable Kafka consumer service in Python that writes micro-batched records to Amazon S3 as Apache Iceberg tables with explicit schema handling
- Added record-level validation with a dead-letter path for malformed events so bad records are isolated without stopping the stream
- Implemented key-based deduplication and replay-safe offset management to keep repeated deliveries from corrupting target tables

**Real-Time Medical IoT Analytics Platform** *Apache Kafka, Apache Flink, Flink SQL, Python, ClickHouse, Grafana*

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process, and score high-frequency medical IoT device data
- Implemented temporal Flink OVER windows partitioned by device ID to compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to handle out-of-order streams
- Designed ClickHouse MergeTree schemas sorted by composite keys to align disk layout with clinical query patterns, with an optimized Python multi-topic batch consumer isolating ingestion